

```
octave:1> X=[2,2,3,4]
X =
```

```
2 2
3 4
```

```
octave:2> det(X)
ans = 2
octave:3> X=[2,2,3,4,3]
X =
```

```
2 2 3
3 4 3
```

```
octave:4> det(X)
error: det: argument must be a square matrix
octave:4>
```

Linear algebra tip: A determinant is defined only on square matrices. If we view a square matrix X , as a linear transformation on a vector A , the determinant of X determines the scale factor of the transformed vector A . That is, if A encloses a finite area in space, the transformed matrix AX will enclose double the area, assuming $\det(X) = 2$.

By any chance, would you remember the case of $\det(X) = 0$? Such a matrix X , is said to be singular. Singular matrices are not invertible. This means that we cannot have an inverse of such a matrix. Let's see what Octave says:

```
octave:6> X=[2,2,2,3]
X =
```

```
2 2
2 3
```

```
octave:9> inv(X)
ans =
```

```
1.5000 -1.0000
-1.0000 1.0000
```

```
octave:10> X=[2,2,2,2]
X =
```

```
2 2
2 2
```

```
octave:11> inv(X)
warning: inverse: matrix singular to machine
```

```
precision, rcond = 0
ans =
```

```
Inf Inf
Inf Inf
```

Utility matrices

Octave also has functions to generate some utility matrices, such as an identity matrix using the command `eye`, a zero-filled matrix using `zeros`, and others. Please consult the Octave software documentation for the detailed treatment of these functions.

The Hilbert matrix

The Hilbert matrix, named after the great German mathematician David Hilbert, is a square matrix with its entries being of the form: $H(i, j) = 1 / (i + j - 1)$. Octave has a built-in function `hilb` to generate Hilbert matrices. For example:

```
octave:16> H=hilb(3)
H =

1.00000 0.50000 0.33333
0.50000 0.33333 0.25000
0.33333 0.25000 0.20000
```

The above is a Hilbert matrix of order 3. Now calculate the determinant of H :

```
octave:18> det(H)
ans = 4.6296e-04
```

Now, calculate $1/\det(H)$:

```
octave:19> 1/det(H)
```

```
ans = 2160.0
```

This is a curious fact. The determinant of a Hilbert matrix is always the reciprocal of an integer. (As stated in the online encyclopaedia of Integer Sequences at www.research.att.com/njas/sequences/A005249)

If you liked that famous matrix, Octave has a few more. Refer to the documentation to play around with them.

Next up

We covered a lot of matrices in the first part of this series on GNU Octave. We will work with more, focusing on solving linear equations, in the next part of this series, after which we will move to other areas of GNU Octave. A few things to explore on your own, besides what we have already talked about, could be:

- Command-line editing in Octave
- Playing with the Octave prompt
- Experimenting with some of your legacy MATLAB scripts in Octave

Happy exploring with GNU Octave and feel free to write in with your queries and suggestions.



Resources

- GNU Octave: www.gnu.org/software/octave
- GNU Octave Function reference: octave.sourceforge.net/doc/index.html
- Wikipedia articles on Determinants, Matrices and Hilbert Matrix

Acknowledgements

Considering the heavy usage of mathematics formulas, LaTeX was an automatic choice to write this article and thanks to the The Not So Short Introduction to LaTeX [<http://ctan.tug.org/tex-archive/info/short/english/short.pdf>] for insights on how to write mathematics well in LaTeX. Hevea [<http://paullac.inria.fr/hevea>] was used for LaTeX-to-HTML conversion and it is awesome.

By: Amit Saha

The author blogs at amitsaha.wordpress.com and he welcomes queries and suggestions at amitsaha.DOT.in@AT.gmail.DOT.com.